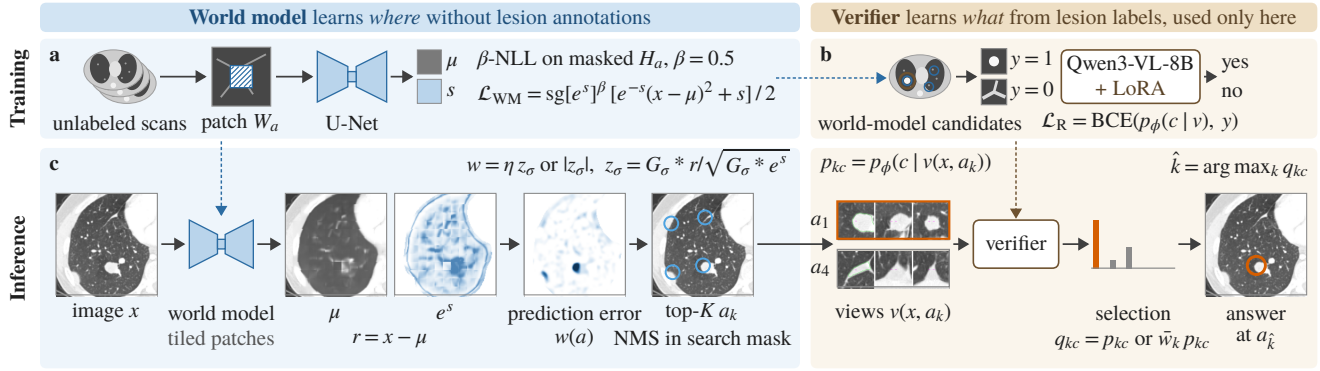

World model learns where without lesion annotations
Verifier learns what from lesion labels, used only here
Training
Inference
a
unlabeled scans
patch W_a
U-Net
μ
s
β-NLL on masked H_a, $\beta = 0.5$
$\mathcal{L}_{\mathrm{WM}} = \mathrm{sg}[e^s]^\beta\,[e^{-s}(x-\mu)^2 + s]\,/\,2$
b
world-model candidates
$y = 1$
$y = 0$
Qwen3-VL-8B
+ LoRA
yes
no
$\mathcal{L}_{\mathrm{R}} = \mathrm{BCE}(p_\phi(c \mid v),\ y)$
c
$w = \eta\, z_\sigma$ or $|z_\sigma|$, $z_\sigma = G_\sigma * r/\sqrt{G_\sigma * e^s}$
image x
world model
tiled patches
μ
$r = x - \mu$
e^s
prediction error
$w(a)$
top-K a_k
NMS in search mask
$p_{kc} = p_\phi(c \mid v(x, a_k))$
a_1
a_4
views $v(x, a_k)$
verifier
selection
$q_{kc} = p_{kc}$ or $\bar{w}_k\, p_{kc}$
$\hat{k} = \arg\max_k q_{kc}$
answer
at $a_{\hat{k}}$